

$$24a^3 + 81b^3$$

ye wala

$$3(8a^3 + 27b^3)$$

$$\cancel{8} y^3 + \frac{1}{\cancel{8}} \cancel{y^3}$$

$$a = y$$

$$b = \frac{1}{2y}$$

$$\left(y + \frac{1}{2b}\right) \left(y^2 - y \times \frac{1}{2b} + \frac{1}{4y^2}\right)$$

$$\left(y + \frac{1}{2b}\right) \left(y^2 - \frac{1}{2} + \frac{1}{4y^2}\right)$$

$$(3a+5b)^3 - (3a-5b)^3$$

$$\frac{(a+b)(a-b)}{a^2-b^2}$$

$$a = \underline{(3a+5b)}$$

$$b = \underline{(3a-5b)}$$

$$a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

$$a^2 + ab$$

$$= [(3a+5b) - (3a-5b)] [(3a+5b)^2 + (3a+5b)(3a-5b) + (3a-5b)^2]$$

$$= \cancel{(3a+5b-3a+5b)} [(9a^2 + \cancel{30ab} + 25b^2) + (9a^2 - \cancel{25b^2}) + b^2]$$

$$= 10b (27a^2 + 25b^2)$$

$$= 270a^2b + 250b^3$$

$$(\cancel{9a^2 - 30ab + 25b^2})$$

$$(a+b)^3 - a^3 - b^3$$

$$(a+b)^3 - (a^3 + b^3)$$

$$(a+b)^3 - [(a+b)(a^2 - ab + b^2)]$$

$$(a^3 + \underline{3a^2b + 3ab^2} + b^3) - a^3 - b^3$$

$$p^3 - (p+1)^3$$

$$p^3 - (p^3 + 3p^2 + 3p + 1)$$